BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Desktop/hw2-matt-b`
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Figure 1: System diagram between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations.

As atmopheric CO2 increases, global mean teamperature also increases due to climate change and the increase of the greenhouse effect, which increases that amount of solar radiation that stays within the atmosphere.

As atmospheric CO2 increases, ocean CO2 also increases because of Henry's law, which states that the amount of CO2 in liquid is propotional to its amount in the gas directly above the liquid. (Source: https://en.wikipedia.org/wiki/Henry%27s_law)

As global mean temperature increases, ocean CO2 decreases because of water solubility. Increased temperature leads to decreased solubility of CO2, decreasing the proportion of CO2 that is stored in the ocean. (Source: https://www.engineeringtoolbox.com/gases-solubility-water-d_1148.html)

This means that there is a dampening feedback between temperature and the ocean carbon cycle (disregarding atmopheric carbon). Increased temperature dampens concentration of oceanic CO2. Considering the three parts as a whole, there are both dampening and amplifying aspects. Increasing atmospheric CO2 increases ocean CO2, while increasing temperature decreases ocean CO2. Henry's law and water solubility push the ocean CO2 concentration in different directions. Overall, however, data shows that oceanic CO2 concentration is increasing over time, suggesting that the impact of Henry's law is more significant (Source: https://ocean.si.edu/conservation/acidification/ocean-acidification-graph).

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

I decided to have two control volumes to account for the two different wastewater discharges. This is important to account for the decay that happens in between the first and second wastewater discharge. If I combined both discharges into one control volume, it would be difficult to account for this decay because there would have to be decay within the control volume.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

```
# Given information
max_dist = 100 #km
iterations = 10000
velocity_river = 10 #km/d
flow_inflow = 250000 #m^3/d
conc_inflow = 0.5/1000 #kg/m^3
flow_d1 = 40000 #m^3/d
conc_d1 = 9/1000 #kg/m^3
flow_inflow_and_d1 = flow_inflow+flow_d1
flow_d2 = 60000 #m^3/d
conc_d2 = 7/1000 #kg/m^3
deposition = 54 #kg/km/d
decay = 0.36 #1/day

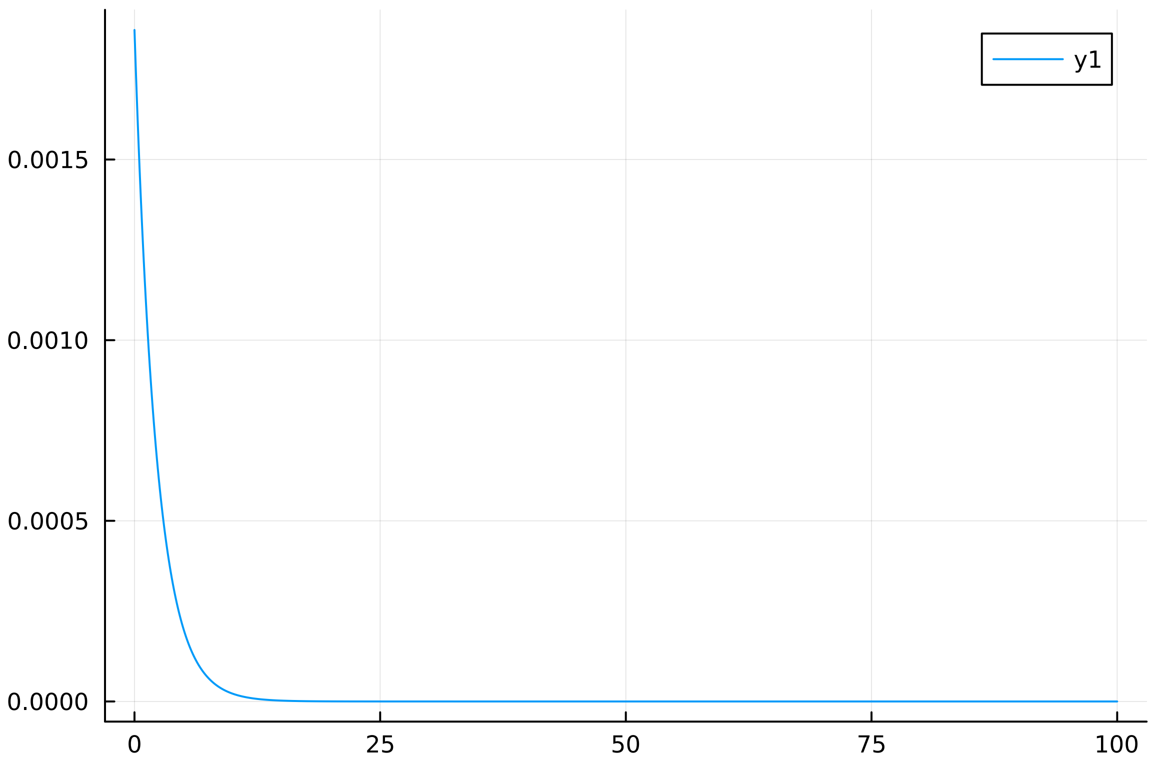
# initialize concentrations with zeros
```

```
concentrations = zeros(iterations)
# set times based on maximum distance and number of iterations
times = range(0, stop=max_dist, length=iterations)
my_array = collect(times)

initial = (conc_inflow*flow_inflow + conc_d1*flow_d1) / (flow_inflow_and_d1)
print(initial)
#concentration = (conc_inflow*flow_inflow + conc_d1*flow_d1) / (flow_inflow_and_d1) + deposi
for i in 1:iterations
    concentrations[i] = (initial + deposition/flow_inflow_and_d1) *(1-decay)^times[i]
end

plot(times, concentrations)
```

```
0.0016724137931034483
```




```
# Given information
max_dist = 100 #km
```

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iterations = 10000
velocity_river = 10 #km/d
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flow_inflow_and_d1 = flow_inflow+flow_d1
flow_d2 = 60000 #m^3/d
conc_d2 = 7/1000 #kg/m^3
deposition = 54 #kg/km/d
decay = 0.36 #1/day

# initialize concentrations with zeros
concentrations = zeros(iterations)
# set times based on maximum distance and number of iterations
times = range(0, stop=max_dist, length=iterations)
my_array = collect(times)

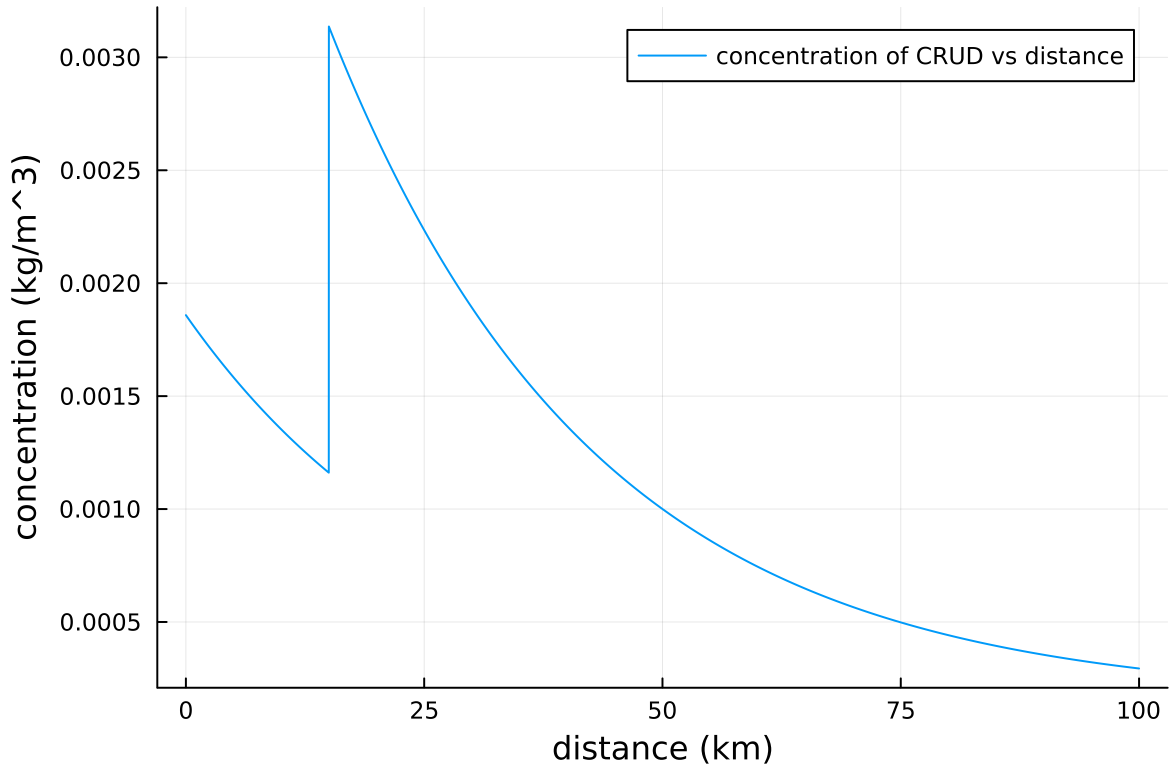
# initial concentration from mass balance with river and first discharge
concentrations[1] = (conc_inflow*flow_inflow + conc_d1*flow_d1) / (flow_inflow_and_d1)
discharged = false

# solve for concentration with forward euler
for i in 2:iterations
    concentrations[i] = concentrations[i-1] - (max_dist/iterations)*(decay*concentrations[i-
    #decay*concentrations[i-1]/velocity_river

    # add second discharge at 15 km
    if times[i]>15.0 && discharged == false
        discharged = true
        concentrations[i] += (concentrations[i-1]*flow_inflow_and_d1 + conc_d2*flow_d2) / (f
    end
end

# add atmospheric deposition
for i in 1:iterations
    if times[i]<15
        concentrations[i] += deposition/flow_inflow_and_d1
    else
        concentrations[i] += deposition/(flow_inflow_and_d1+flow_d2)
    end
end
```

```
# plot results
plot(times, concentrations, label="concentration of CRUD vs distance", xlabel="distance (km)"
```



Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

```
print(maximum(concentrations))
```

```
0.003136522037192094
```

The system is not in compliance with the regulatory limit of 0.0023 kg/m^3. The maximum concentration is around 0.0031 kg/m^3 right after the second discharge.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

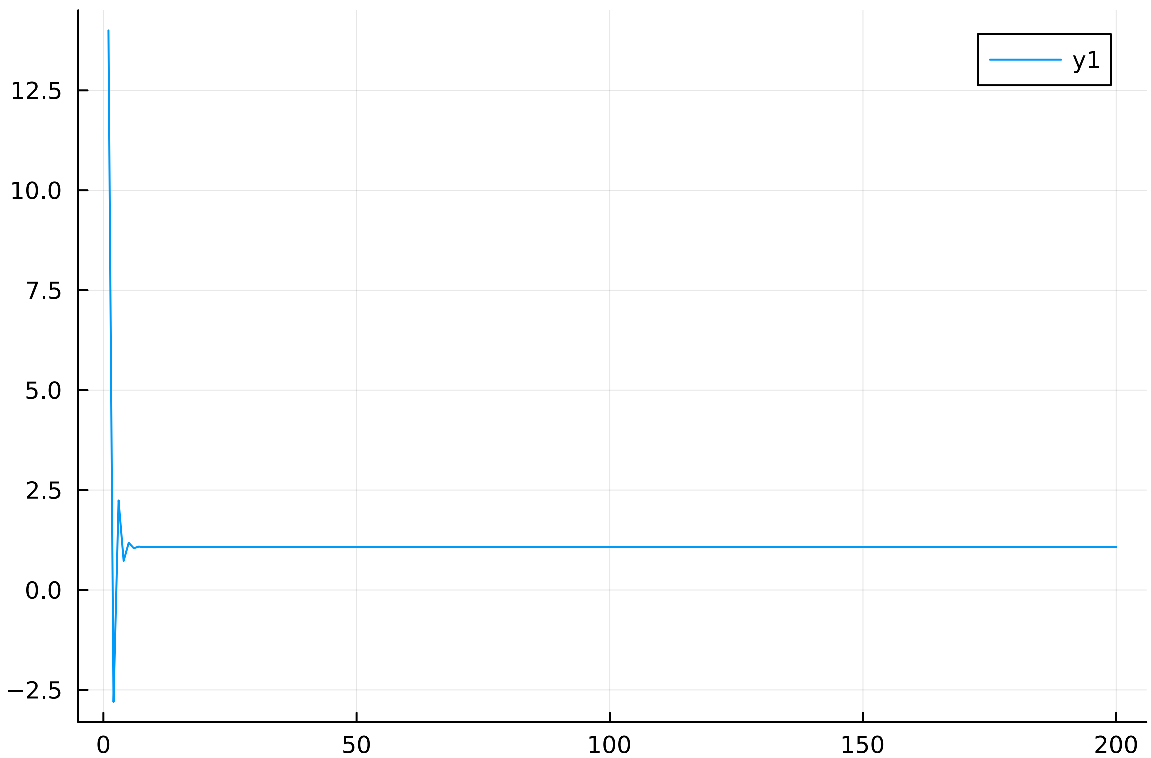
```
years = 200
times = collect(1:years)

S = 1272
A = 221.2
B = -1.3
```

```
a = 0.3

function getTemps(initial)
    T = zeros(200)
    T[1] = initial
    for i in 2:200
        dTdt = (1-a)*S/4 - (A-B*T[i-1])
        T[i] = T[i-1] + dTdt
    end
    return T
end

plot(times, getTemps(14))
```



Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
ai = 0.5
a0 = 0.3
years = 200
times = collect(1:years)

S = 1272
A = 221.2
B = -1.3

function getTemps(initial)
    T = zeros(200)
    T[1] = initial
    for i in 2:200
        if T[i-1] <= -10
            a = ai
        end
        if -10 <= T[i-1] <= 10
            a = ai+(a0-ai)*(T[i-1]+10)/20
        end
        if T[i-1] >= -10
            a = a0
        end

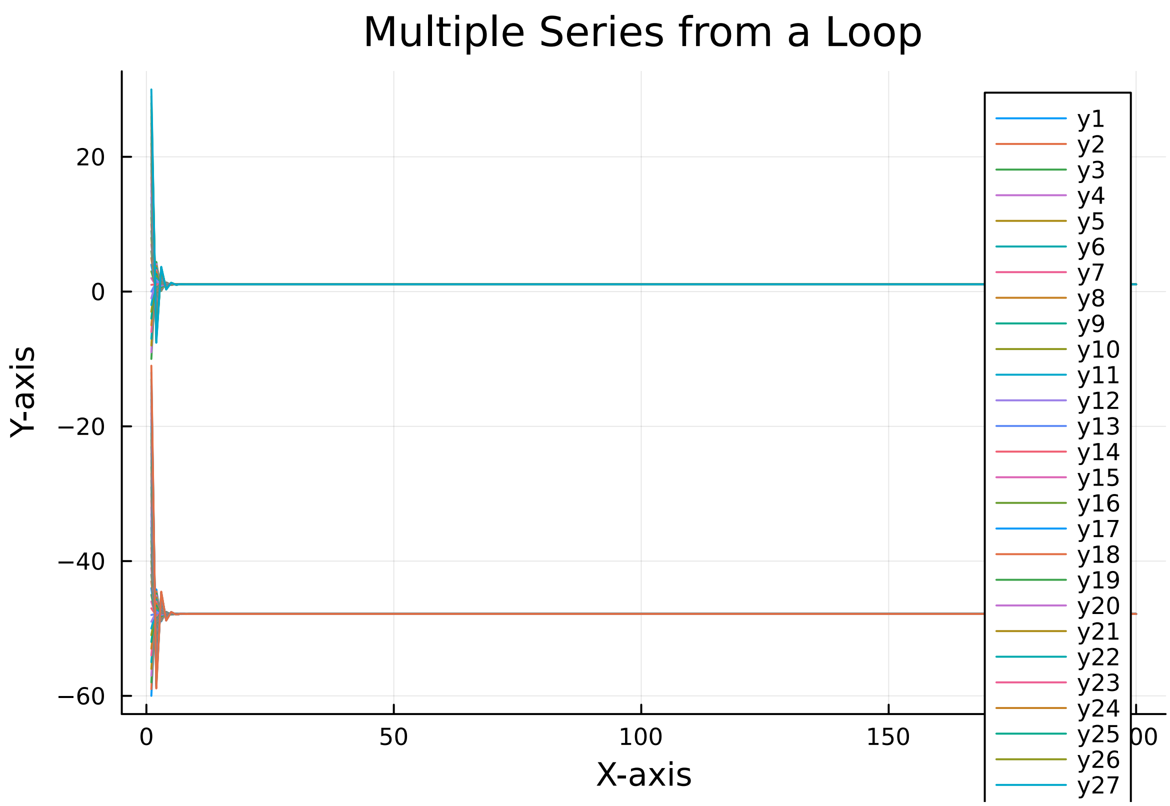
        dTdt = (1-a)*S/4 - (A-B*T[i-1])
        T[i] = T[i-1] + dTdt
    end
    return T
end

p = plot(xlabel="X-axis", ylabel="Y-axis", title="Multiple Series from a Loop")

for i in -60:30
```

```
    plot!(p, times, getTemps(i))
end

p
```



Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

```
ai = 0.5
a0 = 0.3
years = 200
times = collect(1:years)

S = 1368
A = 221.2
```

```
B = -1.3

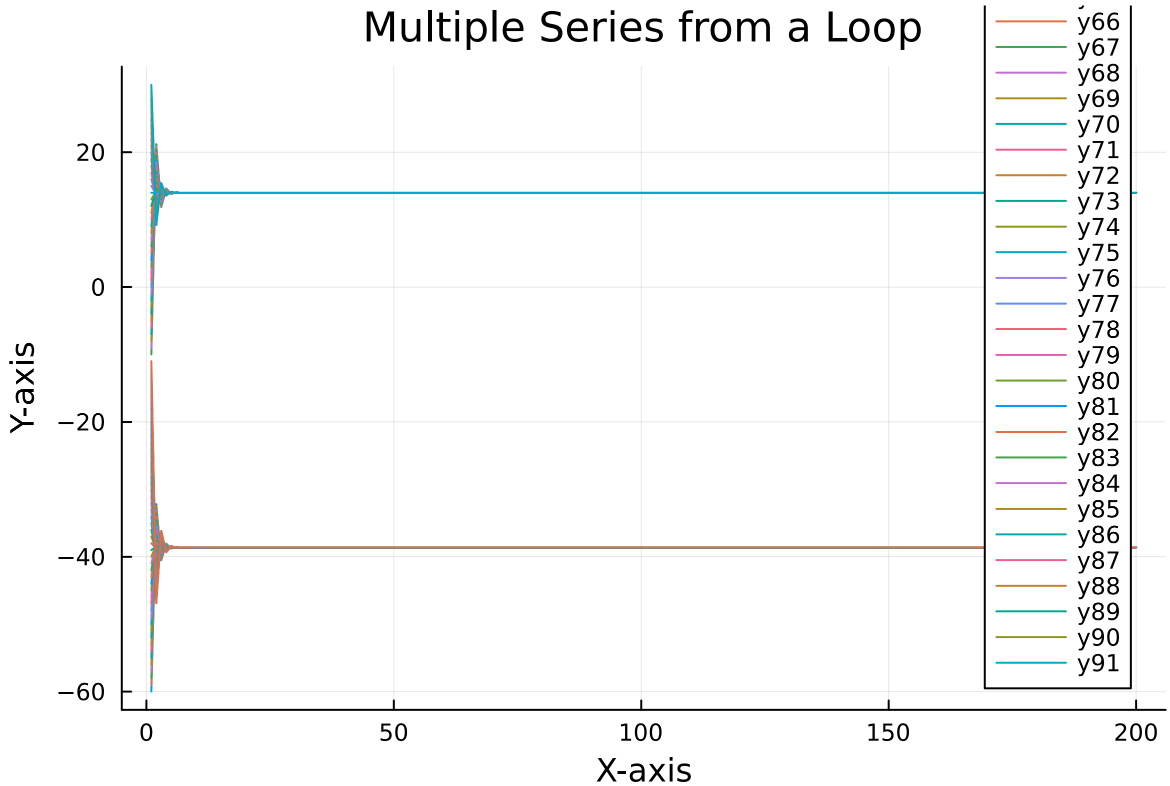
function getTemps(initial)
    T = zeros(200)
    T[1] = initial
    for i in 2:200
        if T[i-1] <= -10
            a = ai
        end
        if -10 <= T[i-1] <= 10
            a = ai+(a0-ai)*(T[i-1]+10)/20
        end
        if T[i-1] >= -10
            a = a0
        end

        dTdt = (1-a)*S/4 - (A-B*T[i-1])
        T[i] = T[i-1] + dTdt
    end
    return T
end

p = plot(xlabel="X-axis", ylabel="Y-axis", title="Multiple Series from a Loop")

for i in -60:30
    plot!(p, times, getTemps(i))
end

p
```



References

List any external references consulted, including classmates.